ASPECT BASED SENTIMENT ANALYSIS MENGGUNAKAN CORRELATION COMPUTING

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OBJECTIVES

1

The system is expected to classify comments based on topics like dirty water, leaks, or meter
issues without manual intervention.

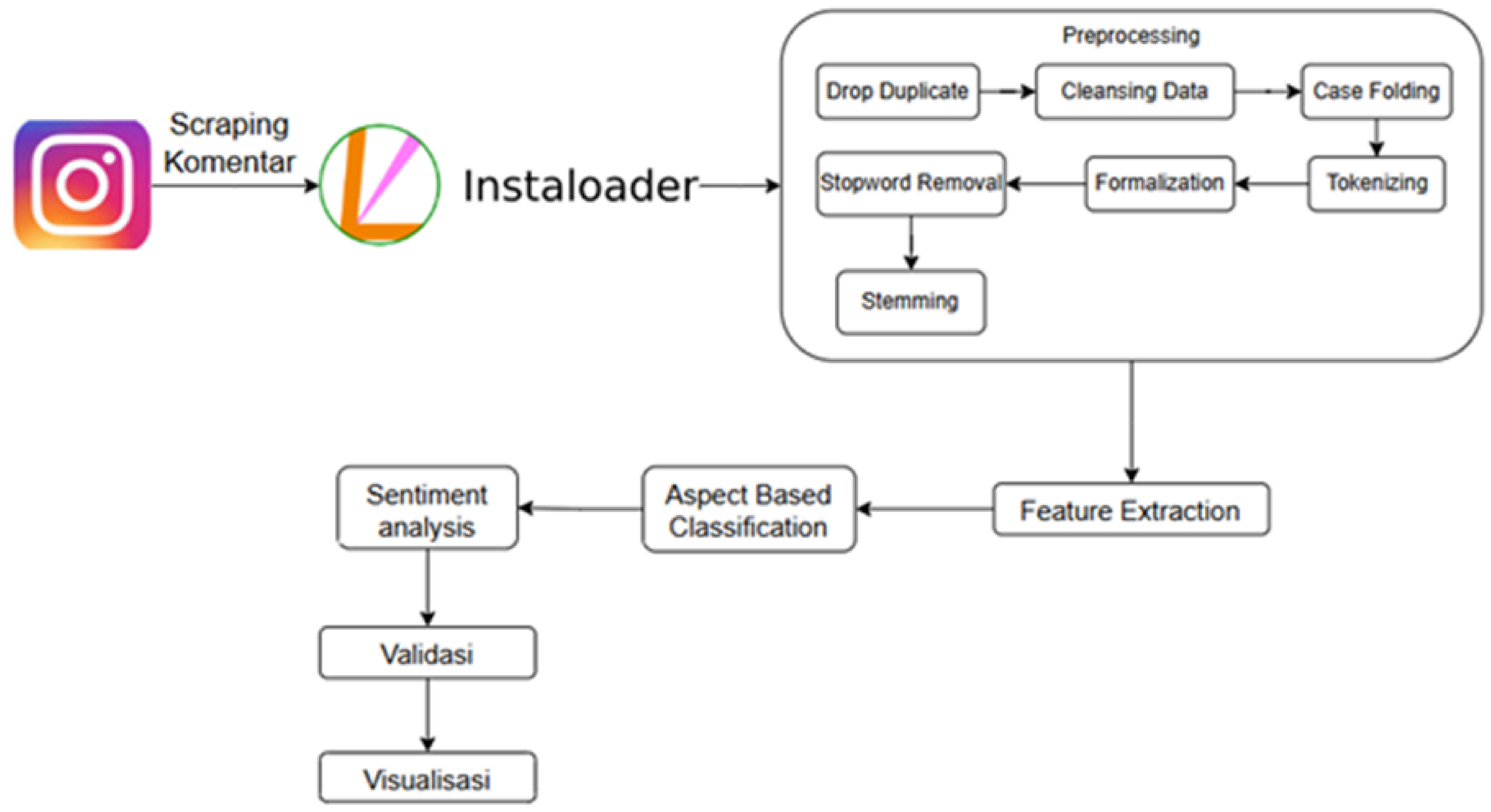
2

It should be able to understand whether comments express positive, negative, or neutral feelings toward PDAM services.

3

By using cosine similarity and rule-based sentiment analysis, the system is expected to perform better for
short, unstructured comment data.

SYSTEM DESIGN



EXPERIMENT SCENARIO

Scenario	Scenario Name	Short Description	Feature
Scenario 1	The ‘Lainnya’ aspect is not ignored and uses TF-IDF for keyword selection in each aspect.	Aspect keyword selection is based on the results of TF-IDF calculations and uses the ‘Lainnya’ aspect to accommodate comments that do not fit into the other 7 aspects.	• Using the ‘Lainnya’ aspect • Using TF-IDF for keyword selection.
Scenario 2	The‘Lainnya’ aspect is not ignored and does not use TF-IDF for keyword selection in each aspect.	Aspect keyword selection is based on the author’s interpretation after reading the entire dataset and uses the ‘Lainnya’ aspect to accommodate comments that do not fit into the other 7 aspects	• Using the “Lainnya” aspect • Not using TF-IDF for keyword selection

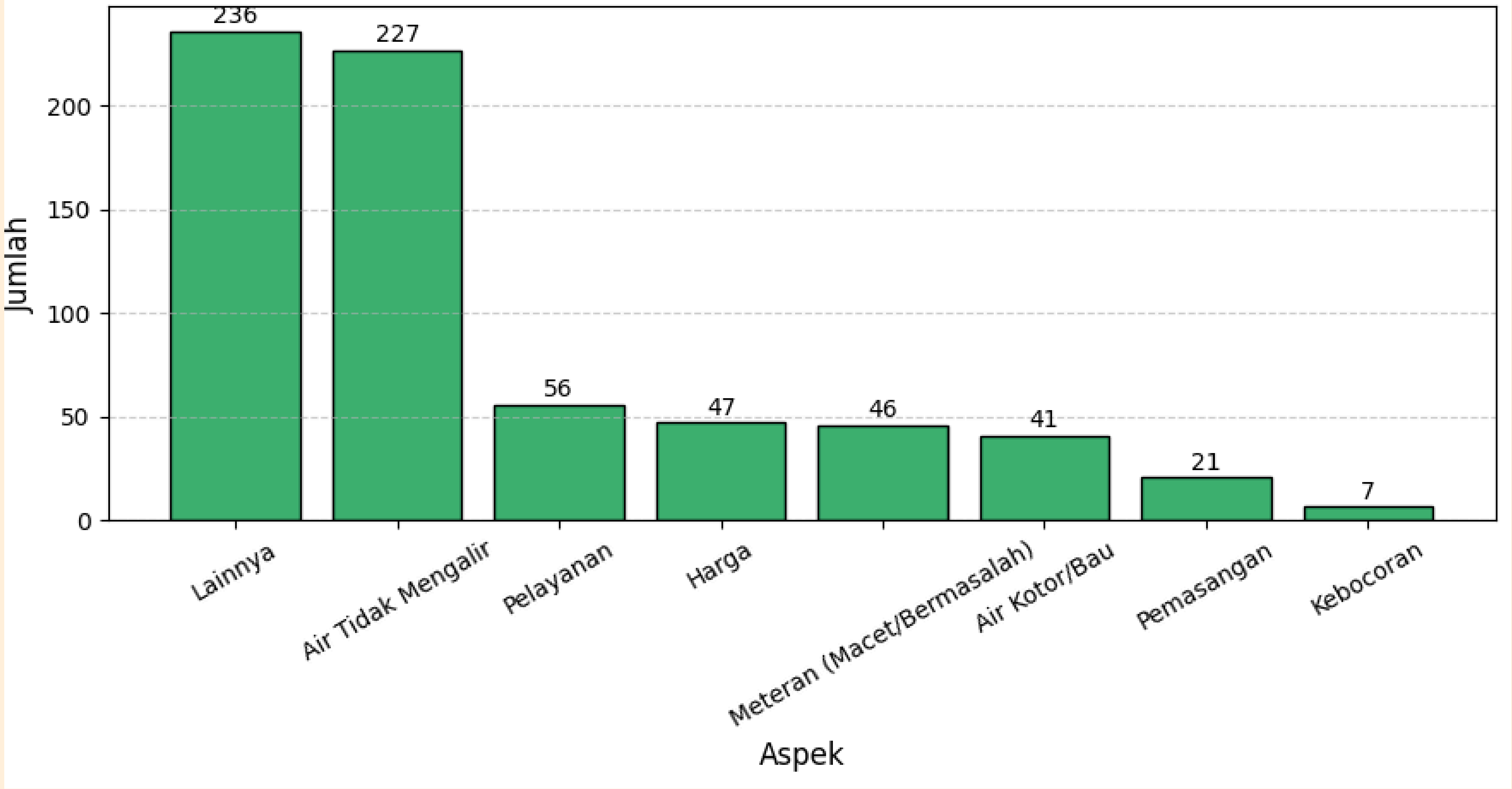
EXPERIMENT SCENARIO

Scenario	Scenario Name	Short Description	Feature
Scenario 3	The ‘Lainnya’ aspect is ignored and TF-IDF is used for keyword selection in each aspect.	Aspect keyword selection is based on the results of TF-IDF calculations, and data with the true ‘Lainnya’ aspect are excluded from the aspect classification process	• Not using the “Lainnya” aspect • Using TF-IDF for keyword selection
Scenario 4	The ‘Lainnya’ aspect is ignored and TF-IDF is not used for keyword selection in each aspect.	Aspect keyword selection is based on the author’s interpretation after reading the entire dataset, and data with the true ‘Lainnya’ aspect are excluded from the aspect classification process.	• Not using the “Lainnya” aspect • Not using TF-IDF for keyword selection

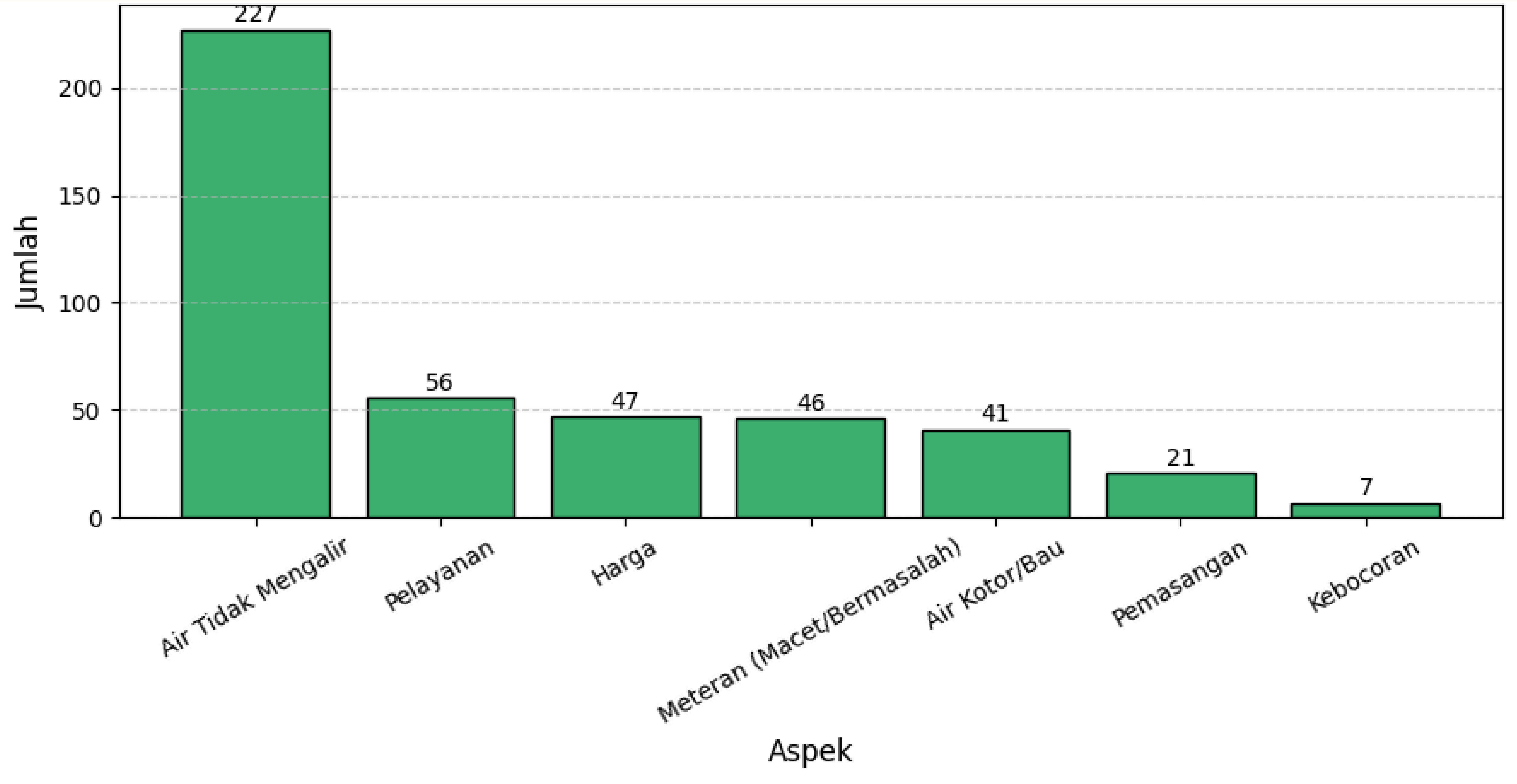
ASPECTS USED

Aspect Name
Air Kotor/Bau
Air Tidak Mengalir
Kebocoran
Pemasangan
Meteran (Macet/Bermasalah)
Pelayanan
Harga
Lainnya

TRUE LABEL DISTRIBUTION OF SCENARIO 1 & 2



TRUE LABEL DISTRIBUTION OF SCENARIO 3 & 4



KEYWORD OF ASPECT

Aspect	Scenario Name			
	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Air Kotor/Bau	["kotor", "bau", "busa", "coklat", "kaporit", "keruh", "kualitas", "warna"]	["kaporit", "kualitas", "buruk", "keruh", "warna", "bau", "kotor", "putih"]	["kotor", "bau", "busa", "coklat", "kaporit", "keruh", "kualitas", "warna"]	["kaporit", "kualitas", "buruk", "keruh", "warna", "bau", "kotor", "putih"]
Air Tidak Mengalir	["macet","mati", "kecil", "keluar", "alir", "debit", "kecil", "padam"]	["dampak", "debit", "tekan", "alir", "keluar", "kecil", "mati"]	[“macet”, “mati”, “kecil”, “keluar”, “alir”, ‘debit”, “padam”]	[“dampak”, “debit”, “macet”, “tekan”, “alir”, “keluar”, “kecil”, “mati”]

KEYWORD OF ASPECT

Aspect	Scenario Name			
	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Kebocoran	["bocor", "resap", "pecah"]	["bocor", "pipa", "pecah", "resap"]	["bocor", "resap", "pecah"]	["bocor", "pipa", "pecah", "resap"]
Pemasangan	["pasang", "instalasi", "sambung"]	["pasang", "sambung", "instalasi", "survey", "aju"]	["pasang", "instalasi", "sambung"]	["pasang", "sambung", "instalasi", "survey", "aju"]

KEYWORD OF ASPECT

Aspect	Scenario Name			
	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Meteran (Macet/Beramasalah)	["pindah", "keran", "rusak", "meter", "remaja", "buram", "embun"]	["pindah", "meter", "patah", "putar", "embun", "buram", "macet", "masalah", "ganti", "rusak"]	["pindah", "keran", "rusak", "meter", "remaja", "buram", "embun"]	["pindah", "meter", "patah", "putar", "embun", "buram", "macet", "masalah", "ganti", "rusak"]
Pelayanan	["pengumuman ", "komplain", "survey", "tanggap", "WhatsApp", "adu", "hubung", "layan", "tindak", "aju", "pemberitahuan ", "respon", "responnya", "slow"]	["whatsapp", "adu", "datang", "tindak", "dadak", "pemberitahuan ", "tanggap", "respon", "update", "pesan", "cepat", "kali", "layan"]	["pengumuman ", "komplain", "survey", "tanggap", "whatsapp", "adu", "hubung", "layan", "tindak", "aju", "pemberitahuan ", "respon", "responnya", "slow"]	["whatsapp", "adu", "datang", "tindak", "dadak", "pemberitahuan ", "tanggap", "respon", "update", "pesan", "cepat", "kali", "layan"]

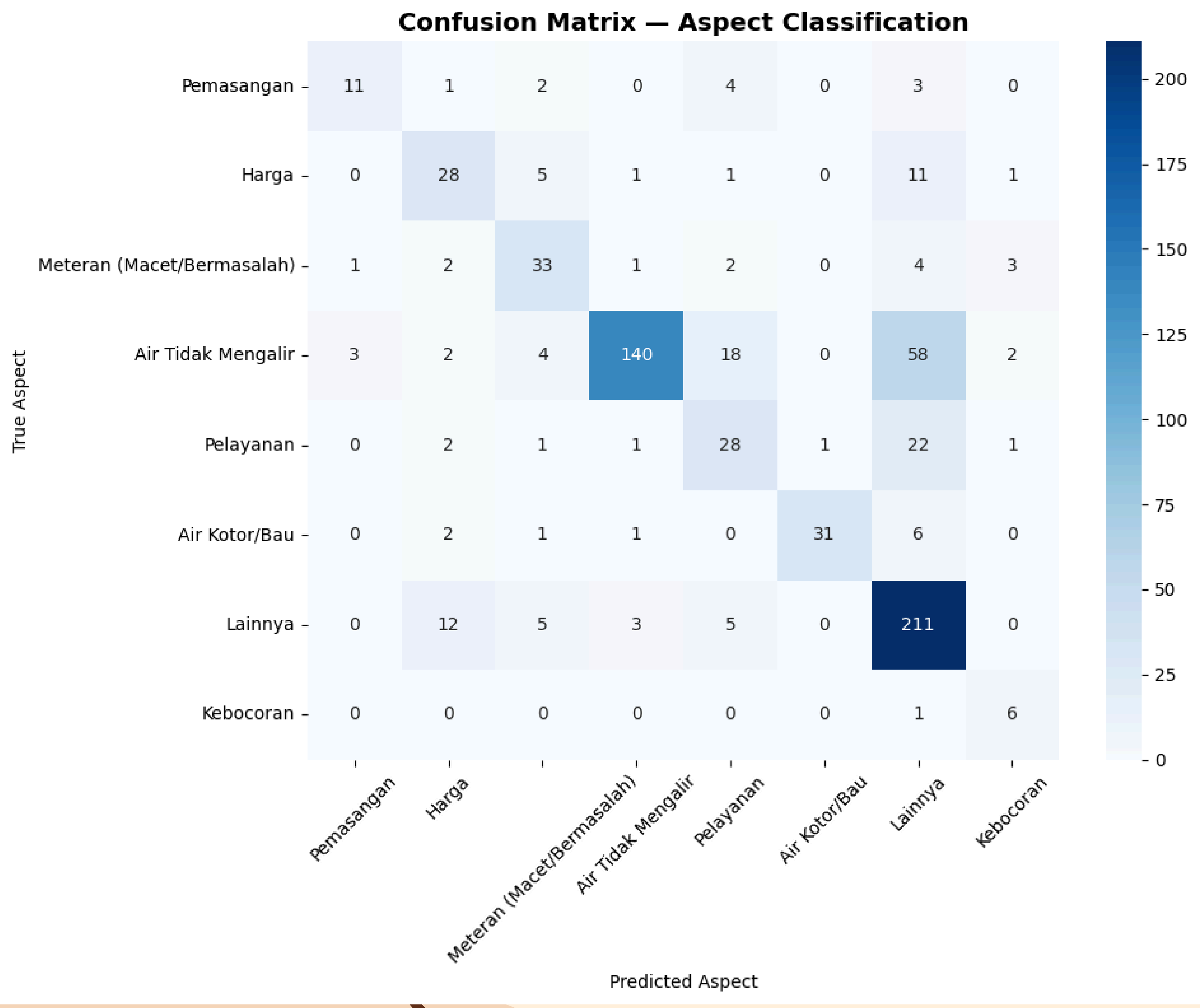
KEYWORD OF ASPECT

Aspect	Scenario Name			
	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Harga	["bayar", "biaya", "denda", "gratis", "harga", "mahal", "murah", "promo", "tarif"]	["bayar", "biaya", "mahal", "murah", "tagih", "rekening", "tarif", "denda", "gratis", "naik"]	["bayar", "biaya", "denda", "gratis", "harga", "mahal", "murah", "promo", "tarif"]	["bayar", "biaya", "mahal", "murah", "tagih", "rekening", "tarif", "denda", "gratis"]
Lainnya	[]	[]		

EXPERIMENT RESULT SCENARIO 1

Aspect	Average Similarity Score
Lainnya	0.00430817
Air Tidak Mengalir	0.121684
Pelayanan	0.150868
Meteran (Macet/Bermasalah)	0.158515
Harga	0.153804
Air Kotor/Bau	0.178864
Pemasangan	0.19009
Kebocoran	0.139842

Total Data	681 data
Accuracy	69.46%
Total Wrong Prediction	208 data
Total Correct Prediction	473 data

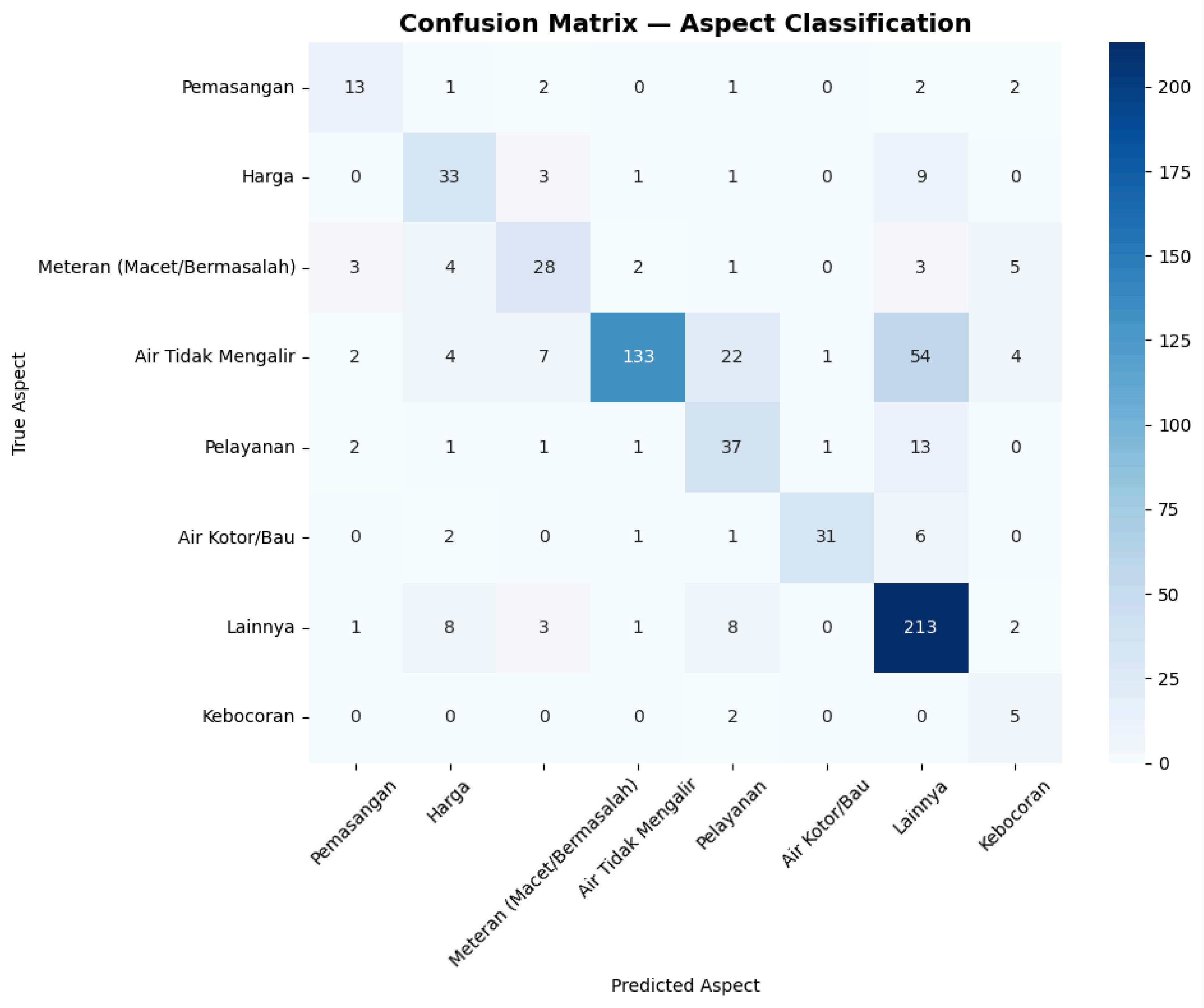


The test results show that the system's accuracy is still relatively low at 69.46%, with 208 data points misclassified. The low accuracy is caused by the small number of words in the comments compared to the aspect keyword dictionary, resulting in low similarity scores. The 'Lainnya' aspect has the lowest similarity score because there are no matching words with the comments. In addition, many comments with the 'Air Tidak Mengalir' aspect are incorrectly classified as 'Lainnya.'

Aspect	Average Similarity Score
Lainnya	0.00404896
Air Tidak Mengalir	0.125877
Pelayanan	0.162739
Meteran (Macet/Bermasalah)	0.151518
Harga	0.152956
Air Kotor/Bau	0.163401
Pemasangan	0.19046
Kebocoran	0.195112

EXPERIMENT RESULT SCENARIO 2

Total Data	681 data
Accuracy	71.07 %
Total Wrong Prediction	197 data
Total Correct Prediction	484 data

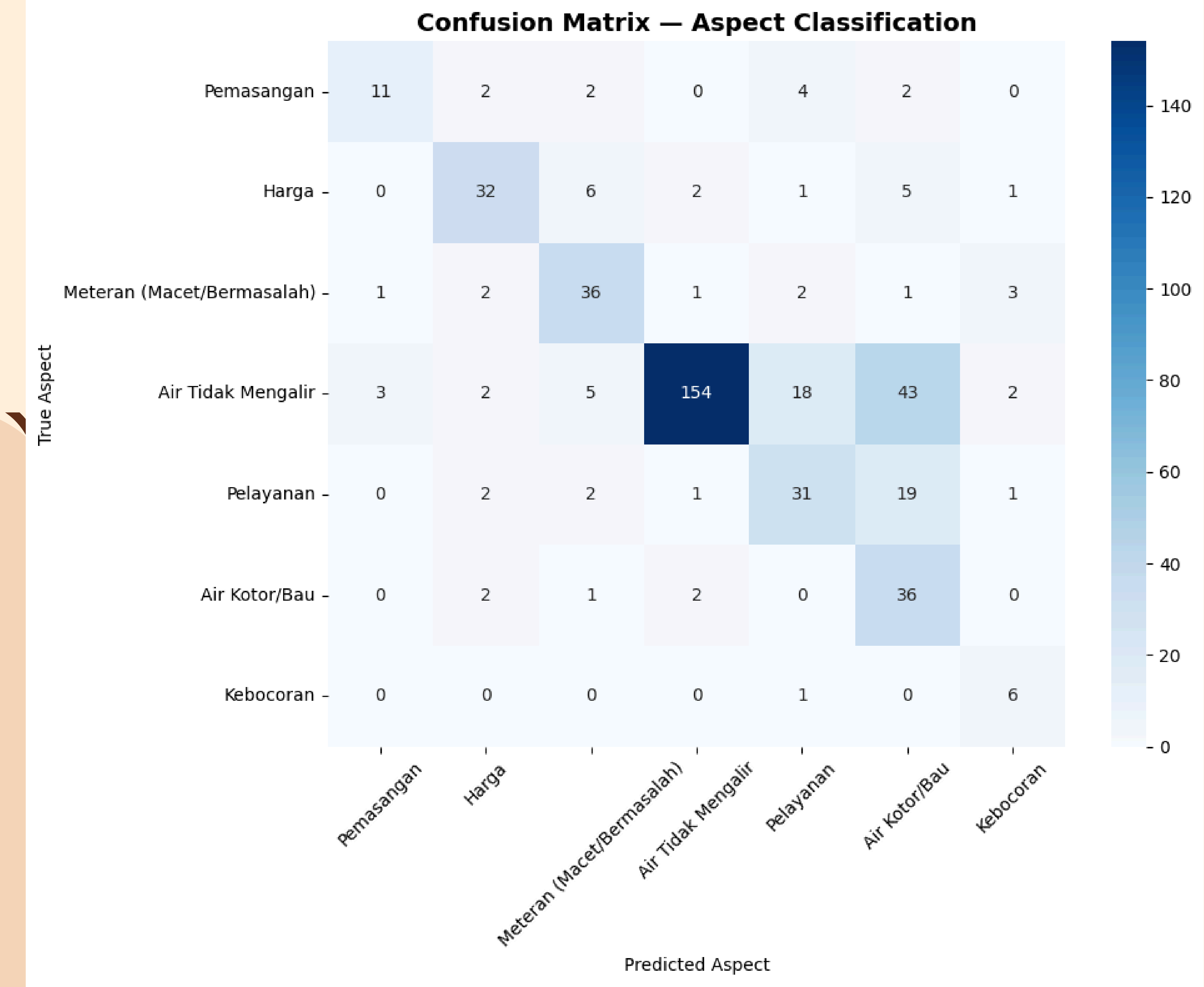


The test results show that the system achieved an accuracy of 71.07%, with 197 data points misclassified. Differences in keyword selection across aspects did not result in a significant improvement due to the similarity of words in the aspect dictionaries. The low accuracy is mainly caused by many comments with the 'Air Tidak Mengalir' aspect being incorrectly classified as 'Lainnya.' This occurs because the words in the comments do not match the keywords of the actual aspect

EXPERIMENT RESULT SCENARIO 3

Aspect	Average Similarity Score
Air Tidak Mengalir	0.103982
Pelayanan	0.142369
Meteran (Macet/Bermasalah)	0.146877
Harga	0.140645
Air Kotor/Bau	0.053464
Pemasangan	0.182235
Kebocoran	0.134092

Total Data	446 data
Accuracy	68.76 %
Total Wrong Prediction	140 data
Total Correct Prediction	306 data

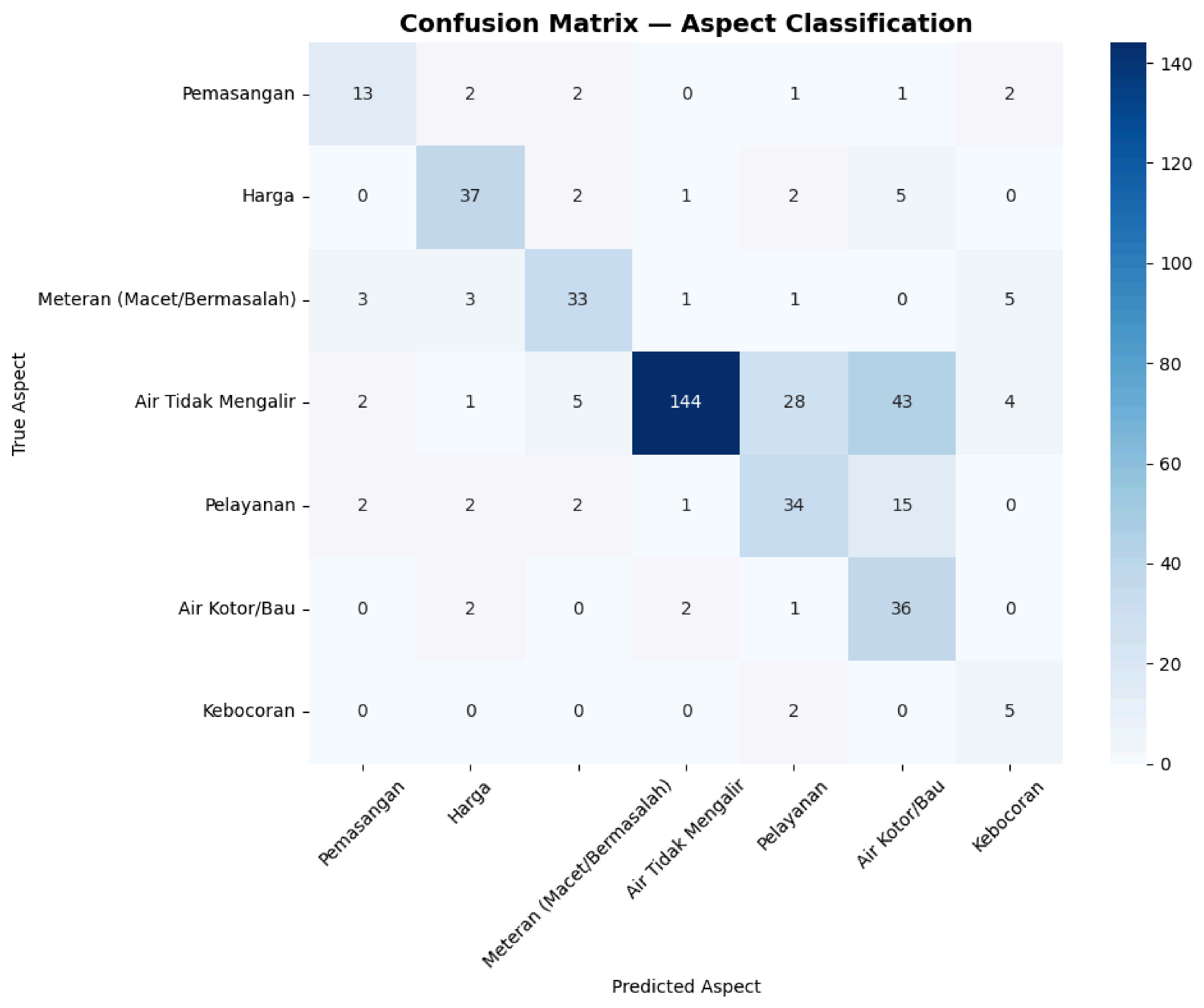


In this scenario, a reduction in data occurs because the ‘Lainnya’ aspect is ignored, resulting in an accuracy of 68.76% with 140 data points misclassified. The similarity score for each aspect is lower than in the previous scenario due to the limited number and variation of keywords in the dictionary, which are unable to cover all word variations in the comments. The low accuracy is also influenced by the small number of words in the comments. Most errors occur in the ‘Air Tidak Mengalir’ aspect, which is often classified as ‘Air Kotor/Bau’.

Aspect	Average Similarity Score
Air Tidak Mengalir	0.101948
Pelayanan	0.142369
Meteran (Macet/Bermasalah)	0.136691
Harga	0.138159
Air Kotor/Bau	0.0528841
Pemasangan	0.182178
Kebocoran	0.194533

EXPERIMENT RESULT SCENARIO 4

Total Data	446 data
Accuracy	67.71 %
Total Wrong Prediction	144 data
Total Correct Prediction	302 data



In this scenario, an accuracy of 67.71% was obtained with 144 data points misclassified. Although there are differences in keyword selection for each aspect, the similarity scores do not show significant changes due to the similarity of keywords in the aspect dictionaries across scenarios. The low accuracy is mainly caused by many comments with the 'Air Tidak Mengalir' aspect being incorrectly classified as 'Air Kotor/Bau.' This occurs due to the lack of matching between the words in the comments and the keywords in the correct aspect dictionary

DONE AND NOT DONE

Done	Not Done
Collecting the dataset	Multi Aspect-Based Classification
Text preprocessing	Sentiment Analysis Model Development
Developing and testing the Aspect-Based Classification model	Testing the Performance of the Sentiment Analysis Model
	Application Development

CHALLANGES

The variety of comment writing styles—such as typos, language differences across comments, abbreviations, and slang makes the formalization process quite time-consuming.

The diversity of words in the comments, the limited number of keywords in the predicted aspect dictionary, and the ambiguous nature of some comments are the main causes of the low accuracy and similarity scores in aspect-based classification.

CONCLUSION

The experimental results show that the cosine similarity-based aspect classification system is able to classify customer comments of PDAM Surya Sembada into the predefined service aspects, although the accuracy remains relatively low (67–71%). The system successfully performs all stages of analysis, from text preprocessing to aspect classification. The low accuracy is caused by the small number of words in the comments, the similarity between aspects, and the limited keywords in the dictionary, particularly between the 'Air Tidak Mengalir' and 'Air Kotor/Bau' aspects. The broadly defined 'Lainnya' aspect also contributes to reduced classification accuracy.

THANK YOU